

2. INTRODUCTION

This chapter identifies the entities, attributes, primary keys, foreign keys, and relationships required for the Etsy Marketplace Database System. These components define the overall structure of the database and provide the foundation for designing a reliable, scalable, and efficient Database Management System (DBMS).

The Etsy marketplace allows independent sellers to list handmade, vintage, and craft products while enabling customers to browse, purchase, review, and track their orders. To manage these activities efficiently, the system stores information in multiple related tables.

2.1 ENTITY IDENTIFICATION

The following entities have been identified for the Etsy Marketplace Database System. Each entity represents a real-world object involved in the marketplace and stores specific information required by the application.

Entity	Description
Customer	Stores buyer information.
Seller	Stores seller information.
Shop	Stores details of each seller's shop.
Product	Stores product information.
Category	Stores product categories.
Order	Stores customer order information.
Order Item	Stores individual products within each order.
Payment	Stores payment information.
Shipping	Stores delivery details.
Review	Stores customer reviews and ratings.
Wishlist	Stores products saved by customers.
Cart	Stores products added to the shopping cart.
Admin	Stores administrator information.

2.2 ENTITY ATTRIBUTES

2.2.1 CUSTOMER TABLE

Attribute	Description
Customer_ID	Unique customer identifier
First_Name	Customer's first name
Last_Name	Customer's last name
Email	Customer email
Password	Login password
Phone Number	Contact number
Address	Delivery address

City	City
State	State
Country	Country
Postal Code	ZIP/PIN code
Registration_Date	Account creation date

2.2.2 SELLER TABLE

Attribute	Description
Seller_ID	Unique seller identifier
Seller_Name	Seller's name
Shop_Name	Shop name
Email	Seller email
Password	Login password
Phone_Number	Contact number
Address	Business address
Rating	Average seller rating
Join_Date	Registration date

2.2.3 SHOP TABLE

Attribute	Description
Shop_ID	Unique shop identifier
Seller_ID	Shop owner
Shop_Name	Name of the shop
Description	Shop description
Logo	Shop logo
Location	Shop location

2.2.4 CATEGORY TABLE

Attribute	Description
Category_ID	Unique category ID
Category_Name	Category name
Description	Category description

2.2.5 PRODUCT TABLE

Attribute	Description
Product_ID	Unique product ID
Seller_ID	Seller who listed the product
Category_ID	Product category
Product_Name	Product name
Description	Product description
Price	Selling price
Quantity	Available stock
Material	Product material
Color	Product color
Size	Product size
Image_URL	Product image
Listing_Date	Product listing date

2.2.6 ORDER TABLE

Attribute	Description
Order_ID	Unique order ID
Customer_ID	Customer placing the order
Order_Date	Date of purchase
Total_Amount	Total bill amount
Order_Status	Pending, Shipped, Delivered
Shipping_Address	Delivery address

2.2.7 ORDER ITEM TABLE

Attribute	Description
Order_Item_ID	Unique order item ID
Order_ID	Order identifier
Product_ID	Purchased product
Quantity	Quantity ordered
Unit_Price	Price per item

2.2.8 PAYMENT TABLE

Attribute	Description
Payment_ID	Unique payment ID
Order_ID	Related order
Payment_Method	UPI, Card, Wallet
Payment_Date	Date of payment
Amount	Paid amount
Payment_Status	Success, Failed, Pending

2.2.9 SHIPPING TABLE

Attribute	Description
Shipping_ID	Unique shipping ID
Order_ID	Related order
Courier_Name	Delivery company
Tracking_Number	Tracking number
Shipping_Date	Shipping date
Delivery_Date	Delivery date
Shipping_Status	Shipping status

2.2.10 REVIEW TABLE

Attribute	Description
Review_ID	Unique review ID
Customer_ID	Customer who reviewed
Product_ID	Reviewed product
Rating	Rating (1–5)
Comment	Review text
Review_Date	Review date

2.2.11 WISHLIST TABLE

Attribute	Description
Wishlist_ID	Unique wishlist ID
Customer_ID	Customer ID
Product_ID	Saved product
Added_Date	Date added

2.2.12 CART TABLE

Attribute	Description
Cart_ID	Unique cart ID
Customer_ID	Customer ID
Product_ID	Product ID
Quantity	Quantity selected
Added_Date	Date added

2.2.13 ADMIN TABLE

Attribute	Description
Admin_ID	Unique admin ID
Admin_Name	Administrator name
Email	Email address
Password	Login password
Role	Administrator role

2.3 PRIMARY KEYS

Entity	Primary Key
Customer	Customer_ID
Seller	Seller_ID
Shop	Shop_ID
Category	Category_ID
Product	Product_ID
Order	Order_ID
Order_Item	Order_Item_ID
Payment	Payment_ID
Shipping	Shipping_ID
Review	Review_ID
Wishlist	Wishlist_ID
Cart	Cart_ID
Admin	Admin_ID

2.4 FOREIGN KEYS

Table	Foreign Key	References
Shop	Seller_ID	Seller
Product	Seller_ID	Seller
Product	Category_ID	Category
Order	Customer_ID	Customer
Order_Item	Order_ID	Order
Order_Item	Product_ID	Product
Payment	Order_ID	Order
Shipping	Order_ID	Order
Review	Customer_ID	Customer
Review	Product_ID	Product
Wishlist	Customer_ID	Customer
Wishlist	Product_ID	Product
Cart	Customer_ID	Customer
Cart	Product_ID	Product

2.5 RELATIONSHIPS

Parent Entity	Relationship	Child Entity
Customer	One-to-Many	Order
Seller	One-to-One	Shop
Seller	One-to-Many	Product
Category	One-to-Many	Product
Order	One-to-Many	Order_Item
Product	One-to-Many	Order_Item
Order	One-to-One	Payment
Order	One-to-One	Shipping
Customer	One-to-Many	Review
Product	One-to-Many	Review
Customer	One-to-Many	Wishlist

Product	One-to-Many	Wishlist
Customer	One-to-Many	Cart
Product	One-to-Many	Cart

2.5.1 RELATIONSHIP EXPLANATION

Relationship	Cardinality	Explanation
Customer → Order	1:M	One customer can place many orders.
Seller → Shop	1:1	One seller owns one shop.
Seller → Product	1:M	One seller can list multiple products.
Category → Product	1:M	One category contains many products.
Order → Order Item	1:M	One order contains multiple products.
Product → Order Item	1:M	One product can appear in many orders.
Order → Payment	1:1	Each order has one payment record.
Order → Shipping	1:1	Each order has one shipping record.
Customer → Review	1:M	One customer can write many reviews.
Product → Review	1:M	One product can receive many reviews.
Customer → Wishlist	1:M	One customer can save multiple products.
Customer → Cart	1:M	One customer can add multiple products to the cart.

2.6 CONCLUSION

The identified entities, attributes, primary keys, foreign keys, and relationships establish the foundation of the Etsy Marketplace Database System. These components organize data efficiently, reduce redundancy, and maintain data integrity. In the next phase, these tables and relationships will be used to create the Entity-Relationship (ER) Diagram, which visually represents the database structure before implementing it in SQL.